

A comprehensive method for measuring flexibility at the individual and global levels: A case Study in the automotive industry

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Abstract— Since the beginning of the 21st century, customer demand has been evolving more and more toward individualized items. In addition, the increase in uncertainties related to geopolitical contexts such as wars, political conflicts, and energy crises, has made the decision process more complex. To respond to these changes, companies are forced to be more flexible, agile, and resilient on the one hand and to apply the concept of mass individualization on the other hand. Knowing that there are several types of flexibility and that higher levels of flexibility generate higher production costs, one of the main challenges for companies is to select the relevant types of flexibility and the appropriate level of them for their projects and objectives.

This paper focuses on modelling, measuring, and determining the appropriate level of flexibility. Since the level of flexibility is mainly related to the level of uncertainties, it is also necessary to model the historical uncertainties and their relationship with the company's objective. For this purpose, the proposed method uses the existing key performance indicators (KPIs) of the company, through which the actual levels of several flexibilities are first measured and then the appropriate levels are defined. Furthermore, by considering the interdependence between flexibilities, an aggregation model transforms the individual measures of flexibility levels into a Global Flexibility Indicator (GFI) that allows decision-makers to prioritize the most important project. In addition, this GFI can be used to model unpredictable events and set flexibility targets based on historical data and events. An application of the proposed method in the automotive industry in the context of a new product introduction is presented.

Keywords: Flexibility indicators, Manufacturing uncertainty, interdependency measure, Aggregation method, decision support system, Automotive industry

I. INTRODUCTION

Uncertainty is a challenging aspect to handle in an industry due to its complexity. Customer demands have become more and more personalized and online purchases increase due to recent restrictions, causing multiple disruptions in the value chain. In today's complex and uncertain environment, these disruptions are in part the consequences of a lack of flexibility and/or resilience in the production system and indicate that production processes are reaching their limits. In addition, the current geopolitical context is generating a strong increase in uncertainty, particularly about material flows and energy supply. Other uncertainties, such as the capacity and morale of human resources after the pandemic as well as financial stability, make the problem even more complex.

Uncertainty has an impact on the entire product life cycle, from the design phase to the recycling phase. Especially for new products, it is already complex to innovate, but it is even more so when supply constraints and uncertainties must be considered in the design phase. In fact, the issues are not just about sourcing and quantity of supply, but also about the flexibility level of all members of the supply chain.

Traditional manufacturing systems are not designed to accommodate this much uncertainty, making it more complex to organize and reconfigure the production system [1]. The need to evolve the current production system to a more flexible, agile, and resilient production system has become a vital need for all companies.

A manufacturing system may have different objectives with regard to flexibility, agility, and resilience. Flexibility generally refers to a system's ability to adjust its operations in response to changing conditions or requirements while minimizing disruption and maintaining performance [2]. Agility, on the other hand, pertains to a system's capacity to respond quickly and effectively to changing conditions or requirements. Resilience refers to a system's ability to recover or bounce back from disruptive events or shocks [3]. [4].

Flexibility can bring about permanent modifications to a system, while a resilient system aims to return to its initial state as quickly as possible. Agility, on the other hand, emphasizes the speed of response. This paper primarily focuses on the aspect of flexibility.

The traditional flexible manufacturing system (FMS) mainly emphasizes the flexibility of the physical part of the production system [5] However, to overcome the aforementioned challenges and achieve a fully flexible system, other aspects of flexibility, such as the flexibility at the cyber level (including IT, decision-making tools and software, workers, managers, back-office employees, etc.) should also be investigated.

With several types of flexibility and increasing costs to change the level of flexibility, it is first necessary to apply a methodology to identify and measure the current levels of flexibility in the production system. Cost-effective determination of flexibility can only be achieved when the different types of flexibility and uncertainty are identified and measured. In addition, flexibility indicators (FIs) are needed for this purpose and could also be used to define the appropriate levels of these based on the objectives and uncertainties in an enterprise.

The objective of this paper is to propose a methodology to help the decision-makers select the appropriate type of flexibility and its relevant levels for manufacturing systems. Section 2 presents the state of the art on the types of flexibility, their interdependencies, and their measurement method. Section 3 explains our proposed seven-step methodology, from scope definition to uncertain event modelling. In addition, several FIs and their measurement methods by using existing company KPIs are developed. Section 4 provides a use case of the Fiat Powertrain Technologies plant in Bourbon-Lancy and explains the application of the proposed methodology in the introduction of a new product in this company. Section 5 concludes with the methodology and its limitations.

II. STATE OF ART

This state-of-the-art focuses on the types of flexibility to understand the definition of each flexibility and the existing method to measure it. In addition, a focus on the aggregation method and the measurement of interdependence is also provided.

A. Flexibility overview

1) Flexibility types

Manufacturing flexibility represents a generic definition of the agility and adaptability of a manufacturing system; however, this term encompasses various variables that influence the behaviour of the overall system and its level. In this context, [6] propose a classification of the different dimensions of flexibility in the manufacturing industry, which is one of the first studies on this topic. They developed eight types of flexibility which served as a basis for subsequent research projects. Other researchers, inspired by this study, have attempted to expand and deepen the subject. [7] added the type of automation, [8] focused on the two dimensions of work and delivery, and [9] also consider the type of new design initiated in the dimensions of flexibility. Recent work by [10] proposes other types of flexibility to address the current environment, supply chain uncertainty, conflict, pandemics, customer demands, and Industry 4.0.

The study of flexibility in the industry has specificities and complexities that attract researchers. By broadening the scope of the study of the concept of flexibility, researchers have attempted to deepen and broaden the fields of study to group the types of flexibility according to the common dimension and purpose in which they operate. The objective was to group/classify the types of flexibility based on correlation, similarity, impact, and importance to the manufacturing system.

In this aim, [11] has proposed three dimensions for type of flexibility:

- *Component or basic flexibilities*: It represents the core component of flexibility by bringing together the machine, material handling, and operations flexibilities that are necessary for the other types of flexibility to function properly. This dimension is more micro-oriented and deals with individual components such as machines or operations.

- *System flexibilities*: it represents a larger scale of work regarding the different characteristics of the manufacturing system, such as the flexibility of process and volume expansion.
- *Aggregated flexibilities*: representing the more generic types of flexibility compared to system flexibilities.

The previous study, [12] adopted the same analogy for their categorization of types of supply chain flexibility by separating the process and logistics into three levels that differ in the scope of their definition.

In some studies, two basic axes for characterizing types of flexibility have been discussed. The first axis includes the decision level and the other concerns the supply chain level. Our previous paper [10] proposes a three-axis classification of flexibilities and considers the physical level, decision levels, and impacts in the company (Fig. 1).

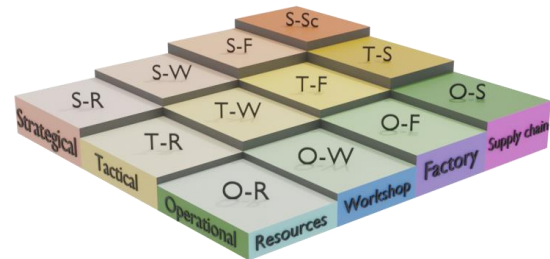


Figure 1: Proposed classification of flexibility matrix

This figure represents the physical level on the right, the decision level on the left, and the business impact on the vertical axis.

The first axis represents the decision levels, from the operational level to the strategic level, the second axis represents the physical levels, from machines and resources to the supply chain and finally, the vertical axis represents the impacts of each class in the value chain. In each class, only certain types of flexibility are classified, and depending on the objective and project of a company, only certain types of them should be considered.

The classification criteria, the types of flexibility in each class, and the identification of the appropriate class for a given problem are discussed in more detail in [10].

2) Flexibility measure method

The ultimate goal of flexibility analysis is to define the precise level of flexibility to make appropriate decisions. Therefore, measuring each type of flexibility is the first step in constructing the overall level of flexibility. [6] reviewed the various indicators proposed in the literature and identify eleven types of flexibility and propose some equations to measure them. For example, the level of product flexibility can be measured by the time or cost required to move from one part to another [6], while [13] uses the number of parts produced per year, while expansion flexibility can be measured by the ratio of the cost of doubling the system's output compared to its initial cost [6], or by measuring the system's capacity to grow. However, this measure is insufficient and must be completed by other key indicators, such as investment capacity which measures the system's capacity to grow. [14] listed various alternatives for measuring market flexibility, citing, as an illustration, the shortage cost of delay in meeting customer orders.

B. Interdependency identification

As presented, the individual measure of flexibility cannot be sufficient to define the strategies of the firm. Thus, it is also necessary to identify the relationships and interdependencies between the different types of flexibility. To the best of our knowledge, the positive or negative interdependency impacts on the decision-making process have not been much discussed in the literature. However, to obtain a result as close to reality, it is essential to consider all interdependencies and propose a method that considers these aspects.

Interdependencies or correlations quantify the relationship between variables. They determine the degree and strangeness to which a variable is affected by changes in other variables. Thus, it is necessary to characterize the level and importance of each type of flexibility on others.

For example, the method, developed by Pearson, [15] measures the linear correlation between two numerical data sets and could also be used for this purpose. It can take values ranging from -1 to 1, while 1 indicates a perfect positive relationship between the two variables, and 0 indicates that there is no linear relationship between the variables. Similarly, the correlation ratio [16] can quantify the relationship between a categorical column and a numerical column. It measures the variance of the mean of the numerical column across the different categories of the categorical column. The correlation ratio, unlike the Pearson and Spearman correlations, cannot measure the "direction" of the correlation; it can only measure its magnitude. A value of 1 indicates that the variance in the numerical data is solely due to variation in the categorical data. A value of 0 indicates that differences in the categorical data do not affect the variance in the numerical data.

Given the nature of flexibility, it is very difficult to use only quantitative methods to model the interdependency of all types of flexibility. In this respect, some existing Multicriteria Decision-Making Methods (MCDM), like AHP (Analytic

hierarchy process) and its simplified version, called Best/Worst (BWM), can be used to solve [17]. This method can be used to evaluate and rank alternative solutions based on a set of criteria. It can also be used to determine the importance of each criterion used to find a solution.

The BWM could be useful in this paper because, in an industrial environment, it is easily understood by all members of a company and the calculated weight could be used as the strangeness of interdependency.

III. METHODOLOGY

As discussed, a comprehensive approach is needed to characterize and quantify organizational flexibility in different aspects on a global scale. Thus, a global methodology in seven steps (Fig. 2), from the identification of the study perimeter and scopes to the modelling of the uncertainty and determination of optimal flexibility level is proposed.

This method does not require the creation of new indicators to measure flexibility, which is one of its advantages. It relies on the subjective correlation of the existing KPIs, already monitored by the organization, with different types of flexibility.

A. Define the scope and perimeter of the study

The first step in increasing the likelihood of project success is to define the project perimeter. These concern the technical aspect (the physical perimeter of the project) or a human aspect (the concerned decision-making level). Moreover, the mapping of a business process management model can be extremely useful. This would be helpful for project framing using IDEF0 methodology. This can model the physical part (machines, ...) and the cyber part (network, decision, ...) of the manufacturing system. This mapping is a resource for the identification of the more relevant uncertainties.

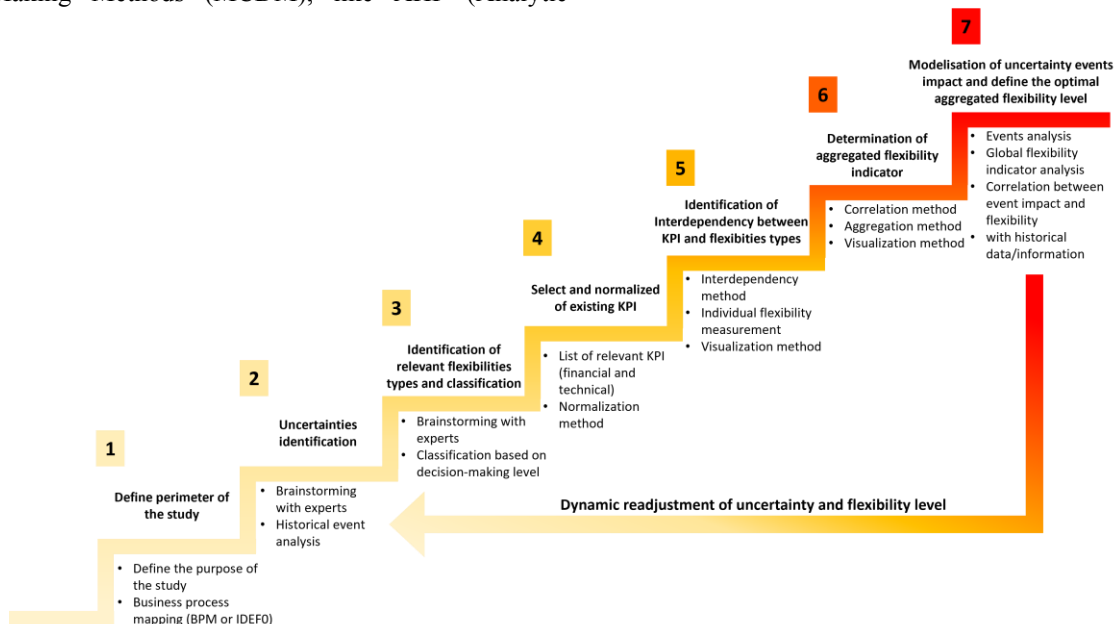


Figure 2: Proposed aggregated flexibility methodology.

The first step of the proposed method is to define the scope of the project. This can be a technical aspect (the physical scope of the project, e.g. introducing a new product or reorganizing of shopfloor) or a human aspect (decision-making). In this aim, business process mapping and management approaches can be extremely useful. For example, scoping the project by using the IDEF0 methodology [18] can allow not only to identify all functions, inputs, outputs, controls, mechanisms, and decisions but also the relationship between several activities. This mapping could also be a resource for identifying the most relevant uncertainties.

B. Uncertainties identification

The second step consists of identifying the most relevant uncertainties in a company. Historical information and knowledge about past uncertain events as well as a brainstorming session with experts can uncover uncertain events that may influence business continuity. By being aware of the possibilities and complexity associated with the structure of the production system, decision-makers can recognize the dangers of downtime and define the appropriate level of flexibility.

C. Identification of relevant flexibility types and classification

Considering many flexibility types, it is necessary to choose a set of appropriate types of flexibility that have a significant impact on strategic projects and objectives. This set of flexibility could be selected when the scope of the study is well defined and the uncertainties that impact the production system have been identified. To do this, it is necessary to understand the business of the company as it is vital to understand which types of flexibility have the most impact on each part of the production system. For this step, brainstorming with specialists who have this understanding would help select the types of flexibility more precisely. Also, as mentioned earlier, a flexibility type without a scope can be difficult to evaluate; therefore, they should be classified using the flexibility type classification matrix. Multi-criteria decision-making methods can still assist in the selection of flexibility types and their classification.

D. Select and normalize existing KPIs

For the fourth phase, due to the complexities of direct flexibility measurement, it is proposed to use the current available KPIs in the company. Thus, it is required to identify the pertinent KPIs based on the perimeter that has been chosen. These KPIs are particularly related to each manufacturing system. The company generally monitors and controls some KPIs as part of its performance controlling and lean manufacturing strategy.

It is essential to select KPIs that cover all technological, financial, human, cyber, safety, and quality considerations. Regarding the nature of each KPI, qualitative or quantitative measurement methods could be used. In this case, to mix several measures, it is necessary to normalize the values, which means removing the units and setting each value to a value between 0 and 1. This will also allow the indicators to be combined in the next steps to produce flexibility indicators. To obtain comparable and normalized values, linearization methods such

as the Linear normalization Max-Min method could be used [19].

E. Measurement of interdependency between KPIs and flexibility types

Once the normalized values of KPIs are calculated, the fifth step consists of transforming them into flexibility indicators. The BWM approach should still be used for each type of chosen flexibility to provide an interdependence factor (α) with the set of specified indicators. Following the determination of the interdependence factor, the flexibility indicators (FI) can be calculated with the normalized value of the KPI. The notations and equation are provided in subsection 3.8.

F. Measurement of aggregated flexibility indicator

By using the individual FI, calculated in the previous step, a Global Flexibility Indicator (GFI) could also be proposed. this step should be performed in the same way to aggregate the FIs and acquire that will be used in the next step 7. Once again, the GFI must indicate the global performance of the manufacturing system. Furthermore, this would help to construct the objective thresholds to achieve depending on the uncertain events. The formulations are provided in subsection

G. Modelisation of uncertainty events and definition of the optimal aggregated flexibility level

The final step of the proposed methodology is the most crucial step from a business viewpoint. Indeed, the FIs and GFI calculation methods consider the uncertain event that occurred in the past. For example, KPIs might be retrieved before, during, and after an uncertain occurrence. The calculation of GFI and FI and consequently the identification of flexibility levels are necessary to identify the performance of a production system in uncertain events. With the provided details by the individual FIs, it would be possible to establish the threshold. The thresholds could be used to prioritize decisions and investments.

H. Method to calculate IFs and GIF

Indices:

i = index of each flexibility type $i \in 1, 2, \dots, I$

j = index of each normalized KPI $j \in 1, 2, \dots, J$

Parameters:

Y = Existing KPI

X = Flexibility type

FI = Flexibility Indicator

GFI = Global Flexibility Indicator

I = Total number of flexibility types.

J = Total number of normalized KPI

α_{ij} = Interdependence factor (between KPI & Flexibility type)

β_i = Interdependence factor (between FI & GFI)

v_j = KPI value

$$FI_{x_i} = \sum_{j=0}^Y \alpha_{ij} \cdot v_j \quad (1)$$

$$GFI = \sum_{i=0}^x \beta_i \sum_{j=0}^y \alpha_{ij} \cdot v_j \quad (2)$$

IV. USE CASE AND RESULT

This study's use case concerns the Fiat Powertrain Technologies Bourbon-Lancy (FPT-BLY) factory in the automotive industry. This plant produces engines for on-road, off-road, and maritime applications. FPT-BLY is a subsidiary of Powertrain Technologies France (FPTF), a subsidiary of the FPT Industrial group that specializes in the design, manufacture, and marketing of heavy engines. All the engines in the Cursor are manufactured by FPT-BLY. The FPT BLY factory has a capacity of approximately 50,000 units per year and over 5700 variants. The five machining shop floors, which house 185 CNC machines, robots, automatic washing machines, and other equipment, produce five major parts (cylinder head, engine block, conrod, camshaft, and crankshaft). The parts are then assembled on the 2 assembly lines which are composed of 258 workstations.

Step 1: The perimeter of this study is defined as the entire FPT-BLY plant. The goal of FPTF is to calculate the GFI of the entire Bourbon-Lancy plant in the context of its strategic target to produce a new family of engines on existing lines. This paper focus on a part of a complete study and KPI values have been anonymized due to confidentiality reasons.

Step 2: Uncertainties were identified in three areas: geopolitics, energy, and supply.

Step 3: Since introducing a new engine is a strategic decision and the scope that has been determined is the FPT-BLY plant, the class S-F (strategic-Factory level) has been selected from the flexibility matrix (Fig. 1).

This case study focuses on 5 of 11 flexibility types chosen by specialists (Table. 1). The experts have chosen the 5 flexibility types based on the previous study [10] in which a complete summary of the several types of flexibility is provided.

TABLE I. Flexibility types for the study

Notation	Flexibility types for the study
X1	Customer
X2	Energy
X3	Supplier
X4	Product
X5	Human polyvalent

Step 4: The KPIs were selected based on the types of FI in the precedent step. Table 2 describes the notation, name, and definition for each existing KPI used in this case.

TABLE II. Selected KPI

Notation	KPI	Definition
Y1	CAPEX	Budget available for investment
Y2	CO2 Emissions	Total CO2 emission
Y3	Supplier Delivery Quality	Component deliver by supplier in quality
Y4	FTQ missing part	Number of non First Time Quality because of missing part
Y5	Efficiency	Value added
Y6	Production cost	Cost of production
Y7	Total Absenteeism	Pourcentage of absenteeism
Y8	Total Inventory Value	Value of inventory

In this step, the linear normalization Max-Min approach is used, which replaces the KPI value with a new value between 0 and 1. Due to confidentiality reasons, the initial values of KPI could not be presented, however, normalized values are presented in Table 3.

TABLE III. Normalize KPI value

Notation	Criteria impact	Value
Y1	Benefit	0,13
Y2	Cost	0,38
Y3	Benefit	0,79
Y4	Cost	0,29
Y5	Benefit	0,33
Y6	Cost	0,23
Y7	Cost	0,88
Y8	Cost	0,24

Step 5: The impact of existing KPIs on flexibility indicators is measured in this step.

The BWM approach is used to define the interdependency factors. For this purpose, the evaluation scores were realized by managers and professionals in the relevant domains. The results are listed in TABLE IV below.

TABLE IV. Interdependency factors

	Y1	Y2	Y3	Y4	Y5	Y6	Y7	Y8
Product	0,06	0,18	0,18	0,12	0,09	0,27	0,02	0,06
Customer	0,16	0,08	0,14	0,16	0,26	0,11	0,05	0,02
Supplier	0,10	0,13	0,30	0,20	0,08	0,10	0,03	0,07
Energy	0,10	0,32	0,07	0,07	0,14	0,14	0,03	0,14
Human polyvalent	0,14	0,03	0,07	0,07	0,14	0,14	0,33	0,07

The interdependency factor (α) is used to calculate the FIs, which are presented in fig. 3.



Figure 3: Actual FIs level

Regarding this step, some conclusions are provided. The plant's flexibility performance is classified and can be visualized. For example, human flexibility is the most efficient, which is not surprising for managers considering. In contrast to other production systems, operators of the assembly line in FPT-BLY follow the product for a quarter of the assembly line (10 to 20 workstations). They must be familiar with all operations for over 800 engine variants. Regardless of the assistance supplied at the workstation, this requires a high level of skill and human flexibility. In the machining line, operators manage a few same types of CNC machines. On the other hand, the prioritized flexibilities can be identified for further investment.

Step 6: The interdependence factor (β) will be used to calculate GFI by using the BWM method. Regarding the study scope for introducing the new product, the global flexibility indicator is calculated as 39.6%.

This indicator provides a reference value for the flexibility of the existing system and enables the tracking of the system's evolution in terms of flexibility performance.

Step 7: Given that step 7 of the proposed methodology is currently under investigation to define the appropriate GFI and FI levels and it will be presented in our next publication.

To summarize this use case, FI has been calculated in the context of introducing a new engine on current manufacturing lines while the plant must continue to also produce the old engines. These FIs provide a new lever to decision-makers and are complementary to the traditional levers of cost, quality, and lead time. The GFI will enable the analysis of unknown events to decide whether the level of flexibility is acceptable and when it must be changed.

V. CONCLUSIONS

The industrial environment, which is complex by an uncertain climate, is beginning to transform as a result of the Industry 4.0 implementation. One of the most essential alternatives to responding to a disruptive event and readjusting the existing organization of the production system comes from the flexibility in various elements of the system. The FMS is well-defined in the literature, and the researcher has largely worked on the physical aspect of the production system. However, it is not well-defined how to implement the FMS in a cyber part of the production system.

This paper proposes a method to identify the appropriate type of flexibility, as well as relevant individual flexibility and global flexibility levels. This method proposes to transform existing Key Performance Indicators (KPIs) into Flexibility Indicators (FIs) to characterize and quantify the level of flexibility. These indicators could be used for a better understanding of the manufacturing system. The proposed method could largely be used in a wide number of strategic projects. This method could also be used to determine the relationship between various types of flexibility. The GFI could also be used to track the flexibility level and performance and trigger modification.

The application of this method in the case of the automotive industry wishes to introduce a new product and highlights the advantages and limits of the proposed method. It was also a good opportunity to determine the actual level of several types of flexibility and to identify the appropriate individual and global levels of flexibility. Another contribution of this study concerns the transformation of existing KPIs into flexibility indicators. Knowing that it's always difficult for decision-makers to characterize and decide about the best type and level of flexibility, these FIs and GFI could be used to simulate historical uncertainties and change the level of FI and GFI, as explained in step seven of the methodology.

Concerning the research perspective, a new method for transforming KPIs into FI that does not rely on empirical data

could also help to produce a more accurate outcome. In addition, to help the decision maker on an investment decision, a mathematical optimization could develop to define the optimal level of each FI for a given budget.

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